

Shu-Wen Yeh

Software Engineering Intern

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SUMMARY

Computer Engineering graduate student at ASU (4.0 GPA) with 5+ years of software engineering experience building enterprise systems at TSMC and Cathay Life Insurance. Passionate about applying my technical expertise while expanding my skills through a Software Engineering Internship (Summer 2026).

SKILLS AND CERTIFICATIONS

Programming Languages: C, C++, Java, Kotlin, Python, TypeScript, JavaScript, SQL, R, Bash, HTML5, CSS

Frameworks & Libraries: Spring Boot, Node.js, Next.js, React, TensorFlow Lite, OpenCV, Flask, Fastify, Cypress, Redux

Tools & Platforms: Git, Docker, Kubernetes, Azure, Linux, MongoDB, MariaDB, MySQL, Prometheus, Grafana, Keycloak

Embedded & Machine Learning: Arduino BLE 33 Sense, TinyML, Edge ML, IMU Sensors, Real-time Inference

Core Competencies: Full-stack Development, System Design, Functional Programming, Object-Oriented Programming, DevOps, Test-Driven Design, Domain-Driven Design, Software Development, Performance Testing, Design Patterns

Certifications: Introduction to TCP/IP - Yonsei University (Coursera, 2026)

EDUCATION

M.S. Computer Engineering, Arizona State University, Tempe, Arizona - GPA 4.00/4.00 Expected May 2027

M.S. Biomedical Informatics, National Yang Ming University (top medical university in Taiwan), Taipei, Taiwan - GPA 3.69/4.00

B.B.A Statistics, National Cheng Kung University (top comprehensive university in Taiwan), Tainan, Taiwan

PROFESSIONAL EXPERIENCE

Senior Engineer

Oct 2021 - Sep 2024

Taiwan Semiconductor Manufacturing Company (tsmc)

Hsinchu, Taiwan

- Established enterprise cloud storage platform serving 30,000+ employees across all fabs on a 4-person team, replacing fragmented legacy file servers within 6 months; received TSMC Special Award for Outstanding Employee (Dec 2022).
- Implemented WebDAV-based services using Spring Boot/Kotlin, React/TypeScript/fp-ts, and Azure CI/CD, enabling in-browser Office editing, version control, and multi-format preview (PDF/Office/Markdown/Video); improved system reliability by 25% and reduced file access latency by 40%.
- Architected distributed carrier management system tracking wafer carriers across TSMC fabs and daily shipments, reducing carrier tracking errors by 30% as part of a 6-person engineering team.
- Built full-stack solution using Spring Boot/Kotlin with coroutines for concurrent processing, Next.js/React/TypeScript, MariaDB, Keycloak SSO; established Azure CI/CD pipelines, Prometheus monitoring, reducing deployment time by 50%.
- Launched training program for 50+ engineers in Functional Programming using TypeScript/fp-ts, reducing code defects by 35%, serving as onboarding mentor and receiving TSMC Distinguished Teacher Award (Dec 2022).

Software Engineer

Jan 2019 - Feb 2021

Cathay Life Insurance

Taipei, Taiwan

- Developed travel insurance system for airport operations with Canvas-based digital signature (jSignature) using Java MVC architecture and Maven build management, improving user enrollment efficiency by 40% and supporting 1,000+ transactions daily, eliminating manual paperwork.
- Engineered AI-powered OCR web service using Python Flask, LSTM, OpenCV, and Tesseract-OCR with multi-threading to automate medical diagnosis document processing, achieving 90% extraction accuracy, reducing manual data entry time by 80%, and cutting processing time by 50% for high-volume medical records workflows.
- Coordinated with programming team to allocate tasks and debug critical technical issues, delivering 6 high-availability insurance claim processing modules on schedule while mentoring 4 junior developers in Java MVC, JSP, and JavaScript.

PROJECTS

Real-Time Posture Detection System using TinyML and IMU Sensors (Arduino Nano 33 BLE Sense)

Fall 2025

Designed and implemented embedded ML to demonstrate expertise in edge computing and real-time data processing.

- Built a real-time posture detection system on Arduino BLE 33 Sense using C++ and TinyML based on ShuffleNet V2, classifying 5 body positions with 82.4% accuracy via IMU sensor fusion (accelerometer, gyroscope, magnetometer).
- Engineered voice-activated LED control system using C++ and TensorFlow Lite (CNN) to recognize 5 distinct commands with 88.48% accuracy.